Product Introduction

ZDH1061 is a VHF/UHF high-gain, high-efficiency power amplifier. It is suitable for a wide supply voltage range of 2.2V to 5.0V. Under typical 4.2V supply conditions, it achieves a P1dB of 33dBm and an efficiency of 52%. The internal input and output impedances are matched to 50Ω, resulting in a very simple external application circuit.

ZDH1061 uses HBT technology and a standard QFN3x3 16-pin package, offering advantages such as low cost and high reliability.

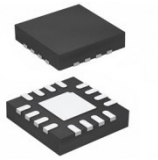
Typical application scenarios

- Analog/Digital Two-Way Radios
- Wireless Data Transmission (FSK, AFSK, OOK, GFSK, MSK)
- Mobile Radio Amplifiers
- Wireless Sensor Networks
- Remote Control and Monitoring
- Consumer Electronics

Absolute Maximum Ratings

Parameter	Value
Storage Temperature	-65°C~+150°C
Operating Temperature	-55°C~+125°C
Supply Voltage (VCC)	+6.0V
Supply Current (ICC)	2.5A
Input Power (RFIN)	+15dBm

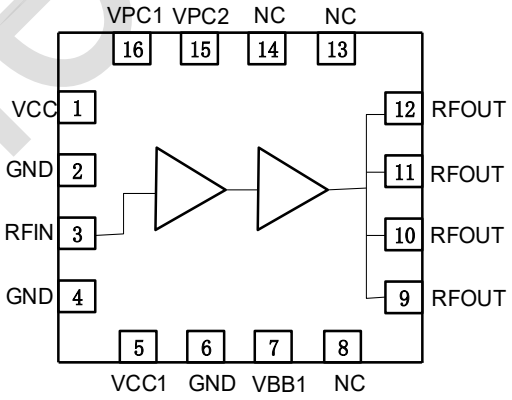
Features



- 2.2V~5.0V single supply voltage, typical current: 1400mA @ 4.2V, 1140mA @ 5V
- Typical gain: 36dB @ 460MHz, VCC=4.2V;
  33dB @ 628MHz、VCC=5V
- Typical P1dB: 33dBm @ 460MHz, VCC=4.2V;
  33dBm @ 628MHz、VCC=5V
- Efficiency: >52%
- Input/output 50Ω impedance matching
- Green lead-free 16-pin QFN3x3 package

This product complies with all relevant regulations and is halogen-free.

Pin diagram (Top View)



PIN No.	Pin Name	Description
1	VCC	VCC Supply Voltage
2,4,6	GND	GND
3	RFIN	RF Input
5	VCC1	VCC1 Supply Voltage
7	VBB1	VBB1 Bias Voltage
9,10,11,12	RFOUT	RF Output
8,13,14	NC	Not Connected. Can be left floating or grounded
15	VPC2	Second Stage Power Control
16	VPC1	First Stage Power Control
17	EPAD	GND

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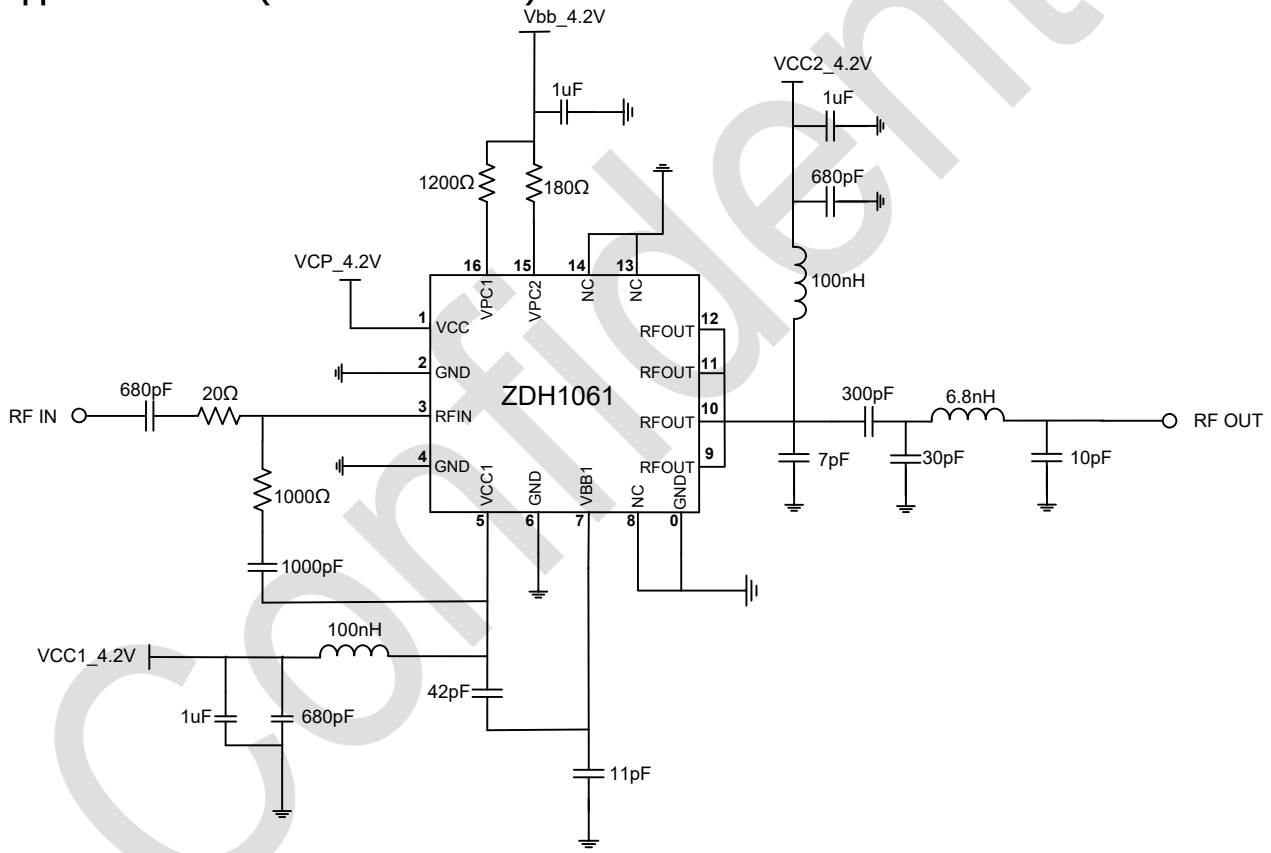
Electrical Parameters

Test Conditions: VCC = +4.2 V, Temp = +25°C, 400MHz~520MHz application circuit, 50Ω test system.

Parameter	Minimum	Typical	Maximum	Unit	Condition
Frequency	400	460	520	MHz	
Gain	-	36	-	dB	
In-band Gain Flatness	-	1.5	-	dB	
Input Return Loss (S11)	-	-8	-	dB	
Output Return Loss (S22)	-	-12	-	dB	
Input/Output Isolation (S12)	-	-47	-	dB	
P1dB	-	33	-	dBm	Measured at 460MHz
Operating Current (Icc)	-	1400	-	mA	
Efficiency	-	52	-	%	

Application circuit (400MHz~520MHz)



Notes:

1. Refurbished PCB material FR4, dielectric constant 4.6, board thickness 0.8mm, 50Ω microstrip line width 1.5mm.
2. 1.8nH wire-wound inductor enameled wire, wire diameter 0.5mm, winding diameter 3mm, 1 turn.
3. PIN8, PIN13, and PIN14 can be left floating or grounded.

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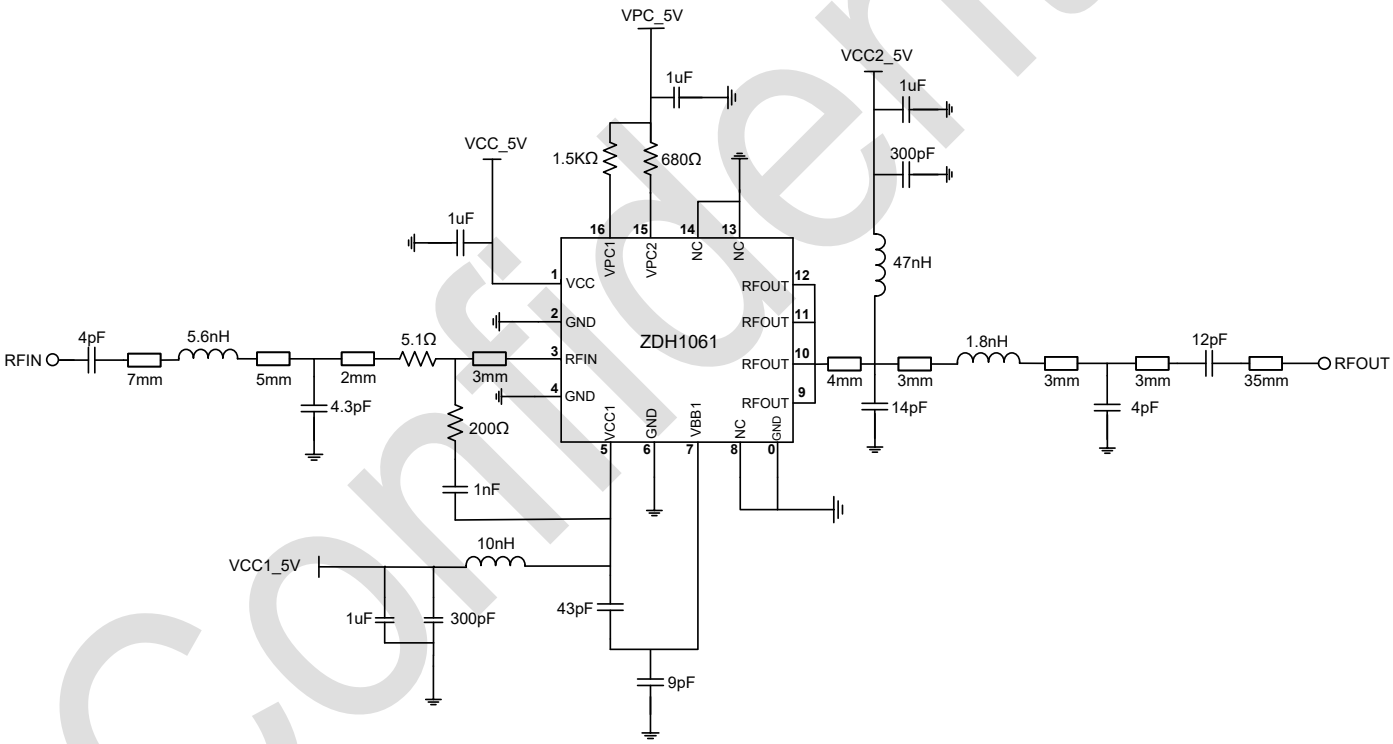
Electrical Parameters

Test Conditions: VCC=+5V, Temp=+25°C, 578MHz~678MHz application circuit, 50Ω test system.

Parameter	Frequency			Unit
Frequency	578	628	678	MHz
Gain	33	33	32	dB
Input Return Loss (S11)	-10	-19	-18	dB
Output Return Loss (S22)	-6	-11	-18	dB
P1dB	33	33	33	dBm
OIP3(1)	44	45	45	dBm
Operating Current (Icc)	1032	1140	1132	mA
Efficiency	44	51	41	%

Note: (1) Two tones, spaced 1MHz apart, each tone has an output power of +18dBm.

Application circuit (578MHz~678MHz)



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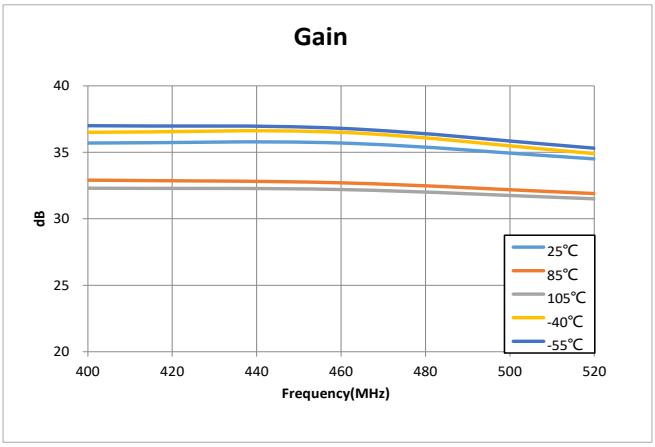
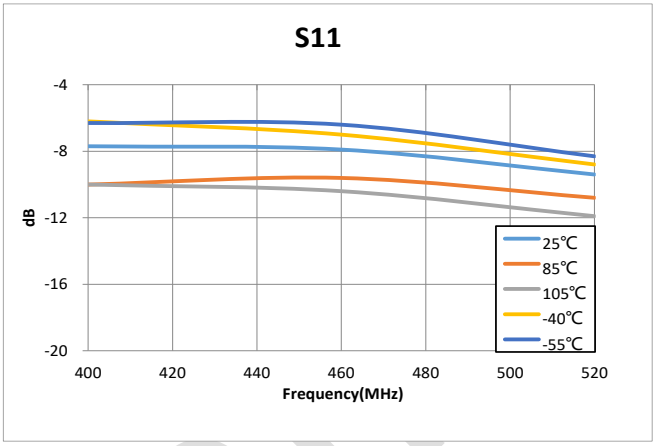
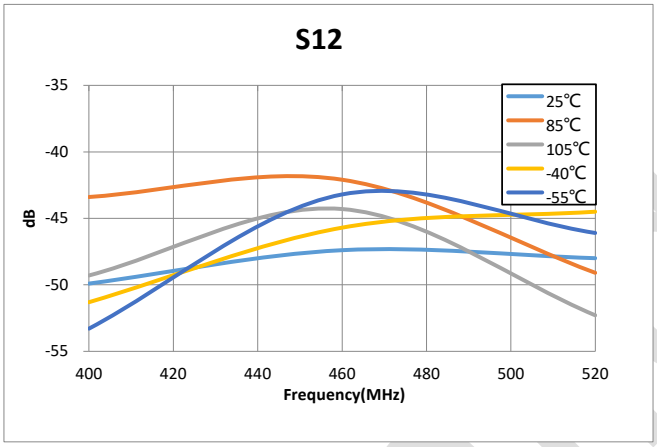
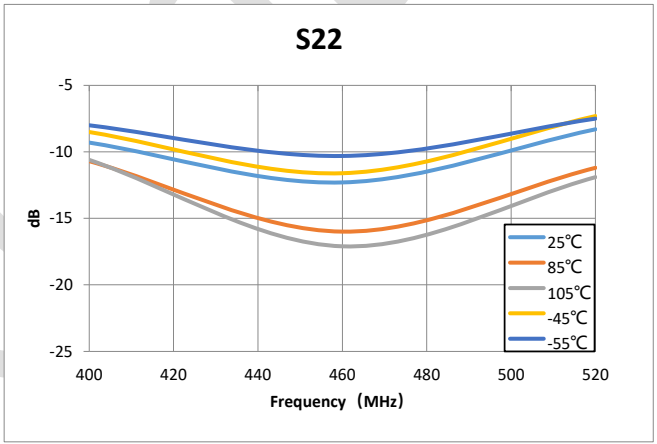
Typical performance curves

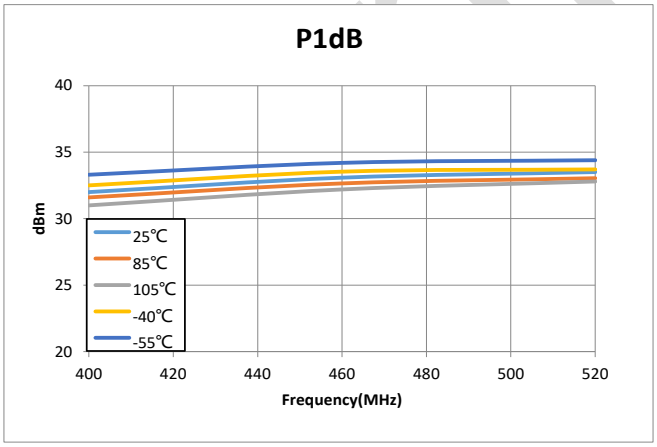
Test conditions: VCC=4.2V, 400MHz~520MHz application circuit, 50Ω test system.

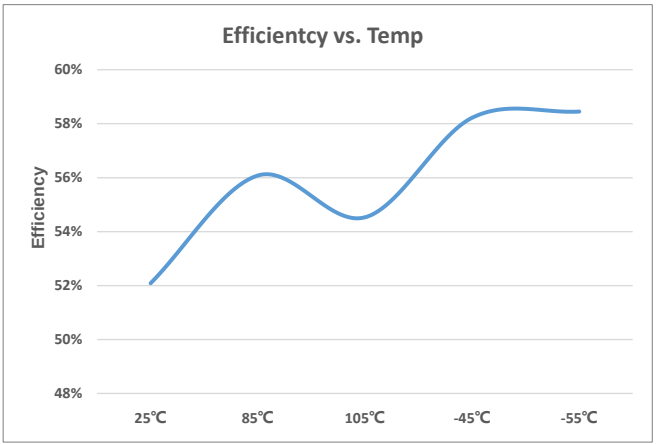






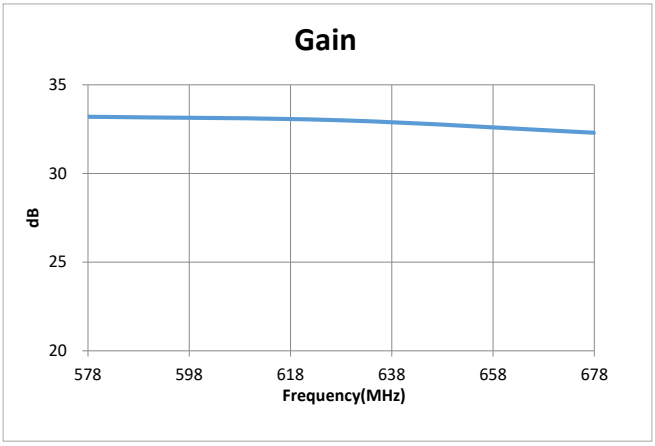


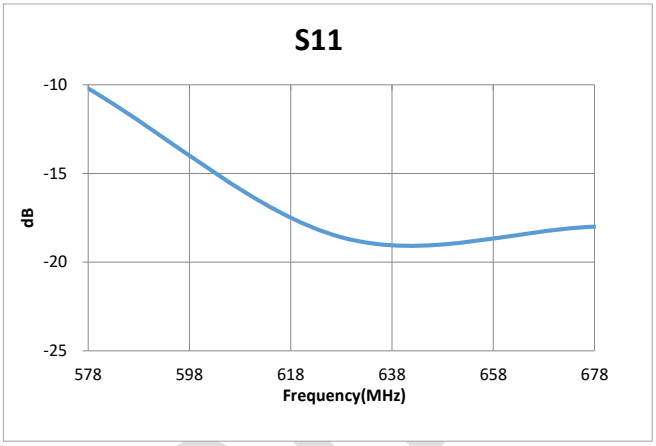


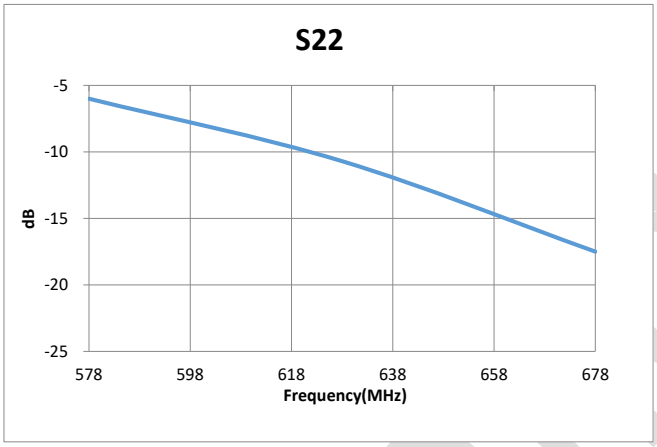


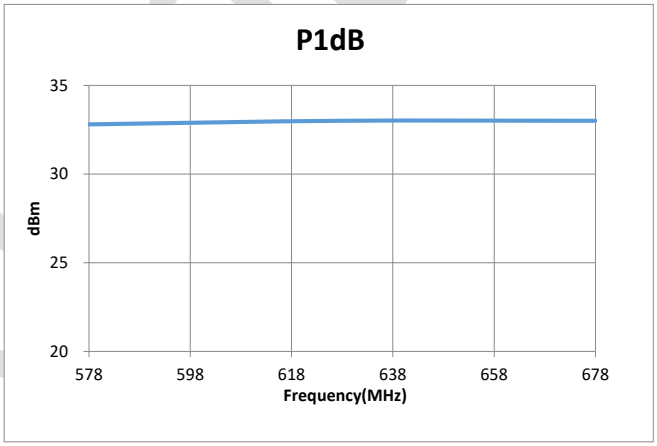
Typical performance curves

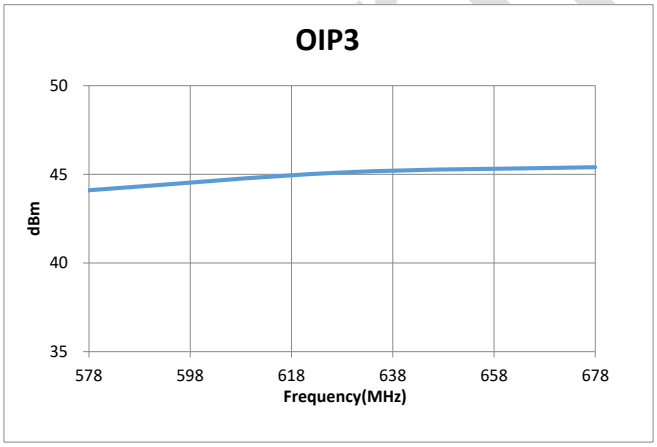
Test conditions: VCC=5V, 578MHz~678MHz application circuit, Temp= +25°C, 50Ω test system.



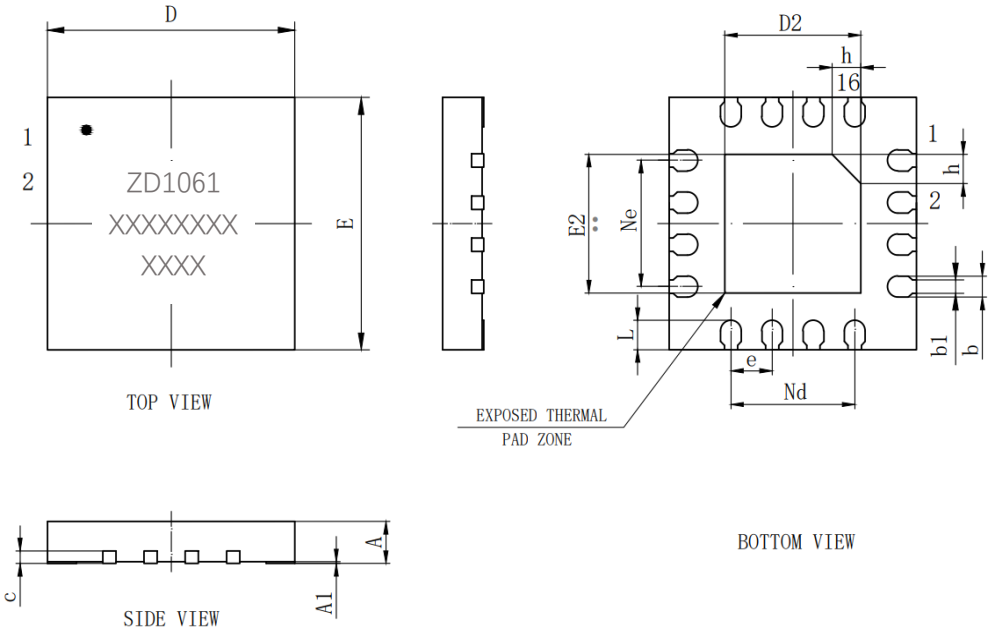








Package size diagram



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	0	0.02	0.05
b	0.18	0.25	0.30
b1	0.11	0.16	0.21
c	0.10	0.15	0.20
D	2.90	3.00	3.10
D2	1.55	1.65	1.75
e	0.50BSC		
Ne	1.50BSC		
Nd	1.50BSC		
E	2.90	3.00	3.10
E2	1.55	1.65	1.75
L	0.30	0.35	0.40
h	0.30	0.35	0.40
L/F载体尺寸	2.05X2.05		

Ordering Information

Model	Marking	Package
ZDH1061	ZD1061	QFN3x3-16

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